

Hw_1

1. Present and discuss the different among Monolithic OS, Layered OS and Microkernel OS?

Monolithic OS : In a monolithic OS, all the OS components, such as file systems, device drivers, and process management, are tightly integrated into a single kernel. These components run in the same address space and share the same memory. It's performance better than others but less flexibility and extensibility.

Layered OS : In a layered OS, the OS is divided into distinct layers, with each layer providing a specific set of services. Each layer communicates only with the layer immediately above and below it, following a strict hierarchy. It easy to add and update service but not efficient as a monolithic OS.

Microkernel OS : A microkernel OS has a minimalistic kernel that provides only essential services like inter-process communication and memory management. Other OS functionalities, such as device drivers and file systems, are implemented as separate user-space processes or servers. High modularity and extensibility let it easier to add and update components. It's safer than others but performance less than others.

In summary, the main differences among these operating system architectures revolve around their design philosophies and trade-offs. Monolithic OSes prioritize efficiency but may suffer from maintainability issues, while Layered OSes provide modularity at the cost of potential performance overhead. Microkernel OSes aim for maximum modularity and reliability but may introduce some performance overhead. The choice of which approach to use depends on the specific requirements and goals of the operating system in question.

2. Explain what are real-time embedded systems?

Real-time systems are a specialized category of computer systems designed to perform specific tasks or functions within a predetermined and stringent time frame. Meaning they must respond to inputs or events within a guaranteed time constraint. And embedded systems are part of a device or a larger system, they are optimized to execute specific functions efficiently and reliably.

3. List Types of Embedded Processor?

Microprocessor

Microcontroller Unit

System-on-Chip

Digital Signal Processor

Field-Programmable Gate Array

Application-Specific Embedded Processor

Graphics Processing Unit

Application-Specific Integrated Circuit

4. Please Lists Models of Embedded Operating System?

FreeRTOS

VxWorks

QNX

Embedded Linux

μ C/OS-II and μ C/OS-III:

INTEGRITY

ThreadX

NuttX

eCos

Windows Embedded Compact

Hw_2

1. Please list the directory below and what information is mainly placed:

/etc/, /boot, /usr/bin, /bin, /usr/sbin, /sbin, /dev, /var/log, /run

/etc : files related to system settings and management. For example, the passwd file that records the account name and the shadow file that projects the password .

/boot : Files that must be read when the system starts, including the system core.

/usr /bin : User-executable command programs, such as find , free , gcc, etc.

/bin : Place various command programs required by the operating system. Such as cp , rpm , kill , tar , mv , rm and ping , etc., and various shells , such as bash , zsh , tcsh , etc.

/usr /bin : User-executable command programs, such as find , free , gcc, etc.

/sbin : Programs that need to be executed to start the system, such as fsck , init and swapon , etc.

/dev : File that stores peripheral device codenames. For example, hard disk /dev/ sda , terminal /dev/tty0 , etc. They actually all point to the peripheral device they represent.

/ var : Store data or temporary archives whose content changes frequently when the system is running. Including user emails, log files recording system activity processes , print job queue files, PID (ProcessID , program identification code) records of system execution programs, etc. Server-specific directories such as the Apache web directory and FTP directory are also located here.

/var/log: Log files. System and application log files are stored here. It contains records of system activities, errors, and other information. Log files help with system monitoring, debugging, and troubleshooting.

/run: Runtime data. This directory holds system runtime data such as process IDs (PIDs) and socket files for running processes. It's often used for communication between processes and managing system services during runtime.

2. When I want to query some attributes of the file /usr/bin/passwd
(1) traditional permissions; (2) file type (3) hidden attributes of the file,
what command can I use to query?

```
ls -l /usr/bin/passwd  
file /usr/bin/passwd  
getfattr -d /usr/bin/passwd or lsattr /usr/bin/passwd
```

3. Write down the Firewall instruction follow by the policy below:

Rule zero: Clear all existing rules (iptables -F...)

Default policy: Except for the custom chain INPUT which is set to DROP, the others are set to ACCEPT by default;

Trust this machine: Since lo is very important to this machine, lo must be set as a trusted device;

Response packet: Allow the machine to actively request and respond to packets that can enter the machine (ESTABLISHED, RELATED)

Trusted User: This is optional if you want to make your host resources available to zone network sources

Rule zero: Clear all existing rules

```
iptables -F  
iptables -X  
iptables -Z
```

Default policy: Except for the custom chain INPUT which is set to DROP, others are set to ACCEPT by default

```
iptables -P INPUT DROP  
iptables -P FORWARD ACCEPT  
iptables -P OUTPUT ACCEPT
```

Trust this machine (allow traffic on the lo interface)

```
iptables -A INPUT -i lo -j ACCEPT
```

Response packet (allow ESTABLISHED and RELATED packets)

```
iptables -A INPUT -i eth0 -m state --state RELATED, ESTABLISHED -j ACCEPT  
iptables -A INPUT -i eth0 -s 192.168.1.0/24 -j ACCEPT
```

```
# Trusted User (optional: allow traffic from trusted users or zone network sources)
# Replace 'trusted_user_IP' with the actual IP address of the trusted user or network
#iptables -A INPUT -i eth0 -s trusted_user_IP -j ACCEPT

# Save the firewall rules
/etc/init.d/iptables save
```

Hw_3

1. Please create a program by yourself. This program can be used to calculate "how many days do you have to celebrate the New Year"?

```
GNU nano 6.2 hwl.sh
#!/bin/bash

current_day=$(date +%Y-%m-%d)
new_year=$(date -d "+1 year" +%Y-01-01)
daysleft=$((($(date -d "$new_year" +%s) - $(date -d "$current_day" +%s))/86400))
echo "Days until New Year: $daysleft days"
```

```
wu@MSI:~$ ./hw1.sh
Days until New Year: 57 days
```

2. Let the user enter a number, and the program can accumulate from one $\times$ two $\times$ three ... until the number entered by the user.

```
wu@MSI: ~
GNU nano 6.2 hw2.sh
#!/bin/bash

# Author : wu
# Copyright(c) Tutorialspoint.com
# Script follows here:

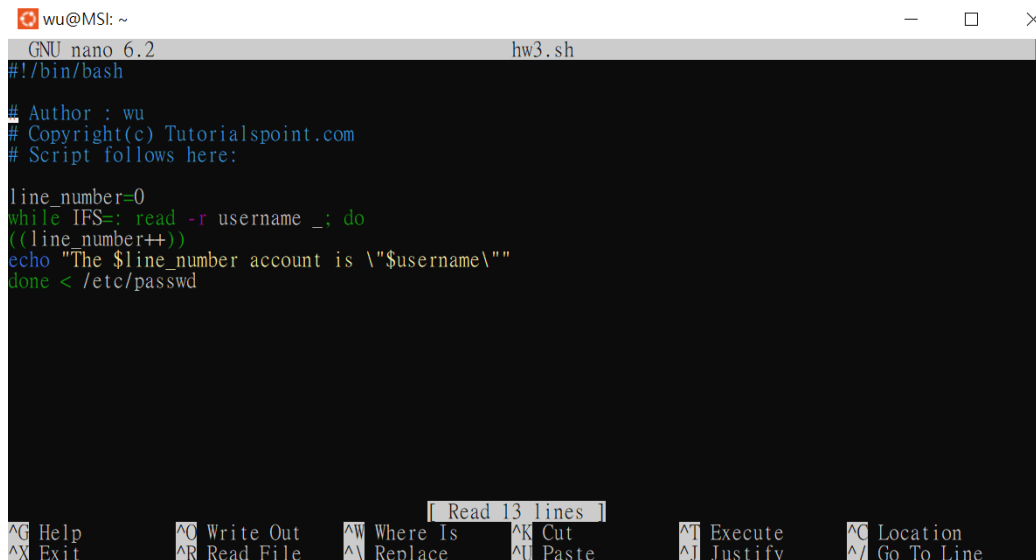
read -p "enter one number to accumulate from 1 multiple to the number : " num
num=$((num))

tem=$((1))
while [ $num -gt 0 ]
do
    tem=$((num*tem))
    num=$((num-1))
done
echo "result : $tem"

Read 19 lines
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
wu@MSI:~$ ./hw2.sh
enter one number to accumulate from 1 multiple to the number : 5
result : 120
```

3. We know that / etc / passwd is separated by : and the first column is the account name. Please write a program that can extract the first column of / etc / passwd, and display each column with a line of string - The 1 account is root , and the 1 represents the line number.

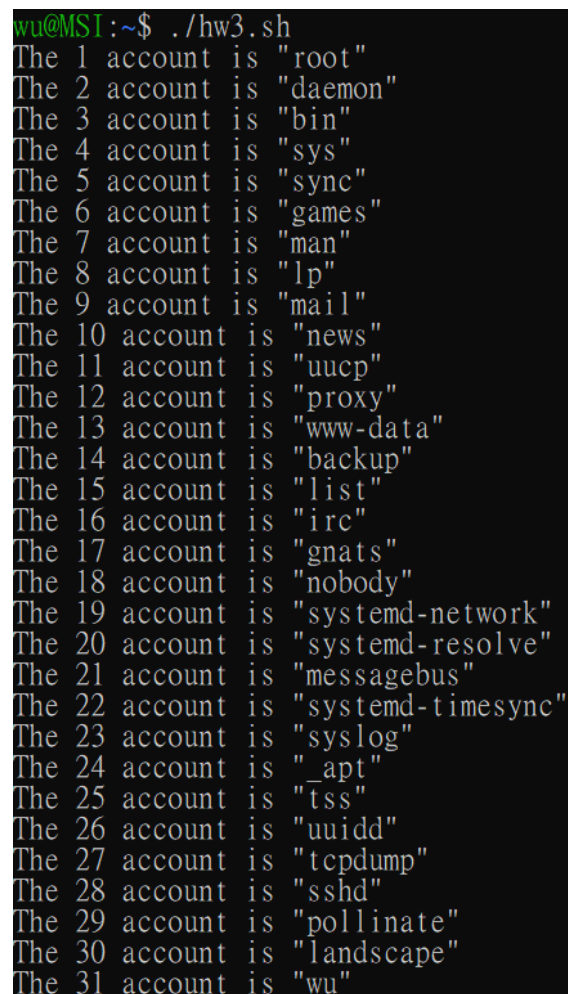


```
wu@MSI: ~
GNU nano 6.2 hw3.sh
#!/bin/bash

# Author : wu
# Copyright(c) Tutorialspoint.com
# Script follows here:

line_number=0
while IFS=: read -r username _; do
  ((line_number++))
  echo "The $line number account is \"$username\""
done < /etc/passwd

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```



```
wu@MSI:~$ ./hw3.sh
The 1 account is "root"
The 2 account is "daemon"
The 3 account is "bin"
The 4 account is "sys"
The 5 account is "sync"
The 6 account is "games"
The 7 account is "man"
The 8 account is "lp"
The 9 account is "mail"
The 10 account is "news"
The 11 account is "uucp"
The 12 account is "proxy"
The 13 account is "www-data"
The 14 account is "backup"
The 15 account is "list"
The 16 account is "irc"
The 17 account is "gnats"
The 18 account is "nobody"
The 19 account is "systemd-network"
The 20 account is "systemd-resolve"
The 21 account is "messagebus"
The 22 account is "systemd-timesync"
The 23 account is "syslog"
The 24 account is "_apt"
The 25 account is "tss"
The 26 account is "uidd"
The 27 account is "tcpdump"
The 28 account is "sshd"
The 29 account is "pollinate"
The 30 account is "landscape"
The 31 account is "wu"
```

Hw_4

When writing a program, its function is:

1. First check whether the name /root/test/logical exists;
2. If it does not exist, create a file and use the "touch" command to create it. After the creation is completed Leave;
3. If it exists, determine whether the name is a file. If it is a file, delete it and create a directory with the file name logical, and then leave;
4. If it exists and the name is a directory, then Remove this directory!

```
GNU nano 6.2 test.sh
#!/bin/bash

path="/root/test/logical"

if [ ! -e "$path" ]; then
touch "$path"
echo "${path} not exist, create a file"
exit 1

elif [ -e "$path" ] && [ -d "$path" ]; then
echo "${path} is exist and is a dir so remove it"
rmdir "$path"
#rm -rf "$path"
exit 1

elif [ -f "$path" ]; then
echo "${path} is exist and is a file so remove it and create dir"
rm "$path"
mkdir "$path"
exit 1

else
echo "not correct work"

fi
```

```
wu@MSI:~$ sudo ./test.sh
/root/test/logical not exist, create a file
wu@MSI:~$ sudo ./test.sh
/root/test/logical is exist and is a file so remove it and create dir
wu@MSI:~$ sudo ./test.sh
/root/test/logical is exist and is a dir so remove it
wu@MSI:~$
```

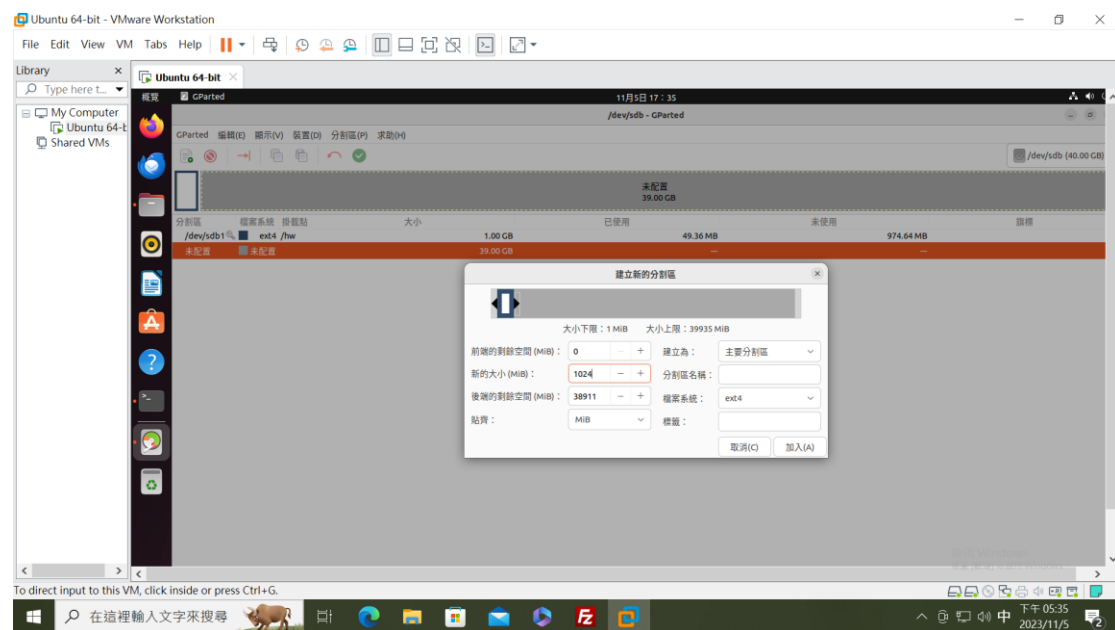

Hw_5

1. Set Up and Configure Disk Quotas : In this exercise, you set up and configure disk quotas and practice using the commands.
2. If I have a Linux host and want to provide mail and WWW services to the outside world in the future, and also plan to provide personal web space services, but I hope that mail will provide 500 MB and WWW will provide 200 MB of space, then how should I plan my host?
3. Utilities timeshift, gzip/gunzip, bzip2, and tar as archiving tools to back up files and user and group files.

Use gparted or instruction to create new partition like

```
# cat /etc/fstab
```

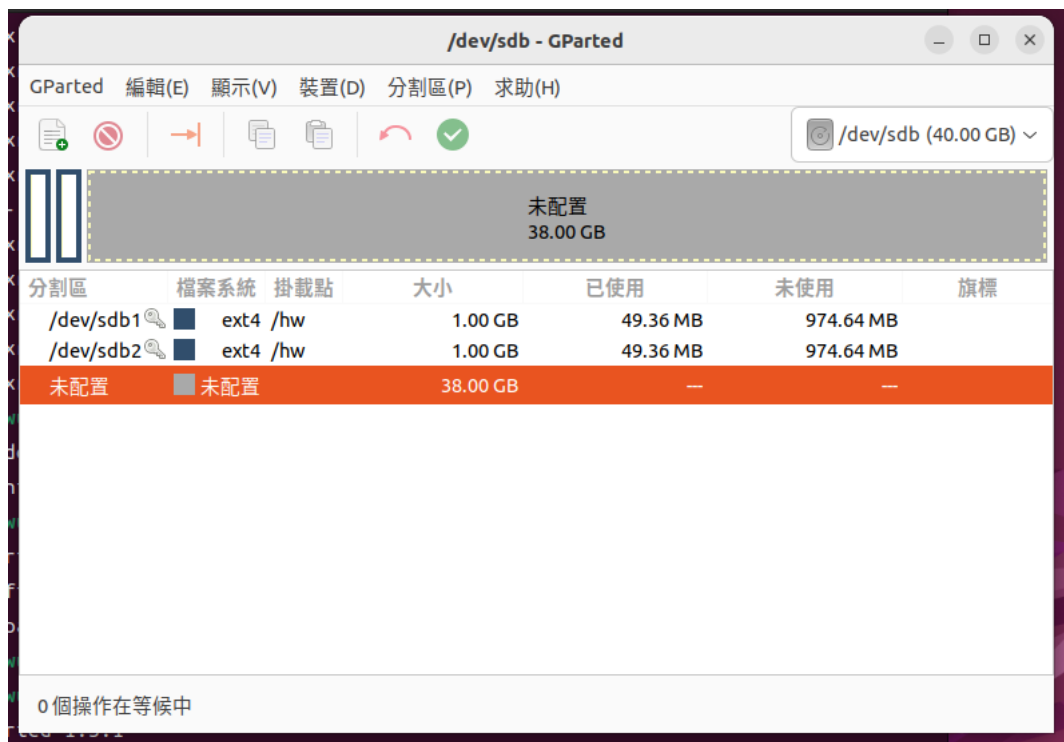
```
LABEL=/home    /home    ext2    defaults,usrquota,grpquota 1 2
```



mount new partition to os directory

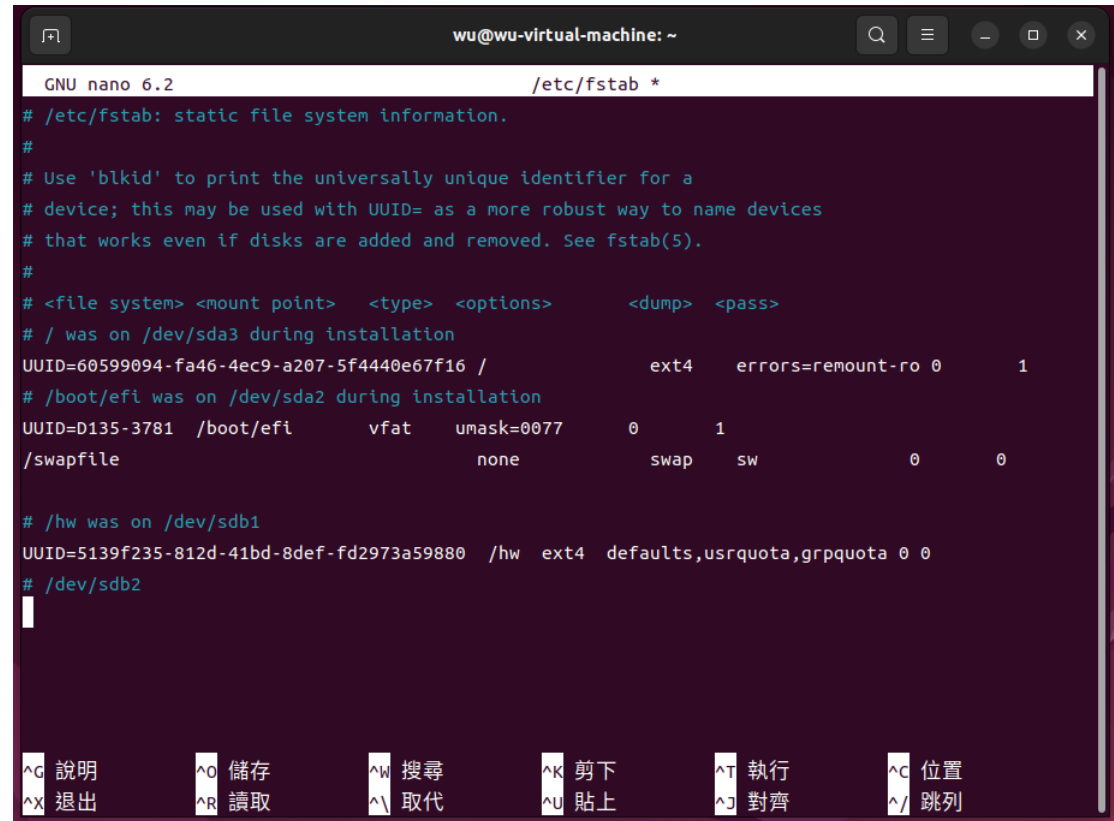
```
sudo mount /dev/sdXY /mnt/new_partition
```

```
wu@wu-virtual-machine:~$ sudo mount /dev/sdb1 /hw
[sudo] wu 的密碼：
mount: /hw: /dev/sdb1 already mounted on /hw.
wu@wu-virtual-machine:~$ gparted
GParted 1.3.1
configuration --enable-libparted-dmraid --enable-online-resize
libparted 3.4
wu@wu-virtual-machine:~$ sudo mount /dev/sdb2 /hw
wu@wu-virtual-machine:~$
```



if after mount UUID not in fstab, add it by manual

```
wu@wu-virtual-machine:~$ sudo nano /etc/fstab
wu@wu-virtual-machine:~$ sudo blkid /dev/sdb2
/dev/sdb2: UUID="62f6f229-12dd-44ee-b777-6418d65a7123" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="b2ec
eb7b-e540-4bfe-a858-2c6fcac7527f"
```



```
GNU nano 6.2 /etc/fstab *
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda3 during installation
UUID=60599094-fa46-4ec9-a207-5f4440e67f16 / ext4 errors=remount-ro 0 1
# /boot/efi was on /dev/sda2 during installation
UUID=D135-3781 /boot/efi vfat umask=0077 0 1
/swapfile none swap sw 0 0

# /hw was on /dev/sdb1
UUID=5139f235-812d-41bd-8def-fd2973a59880 /hw ext4 defaults,usrquota,grpquota 0 0
# /dev/sdb2

```

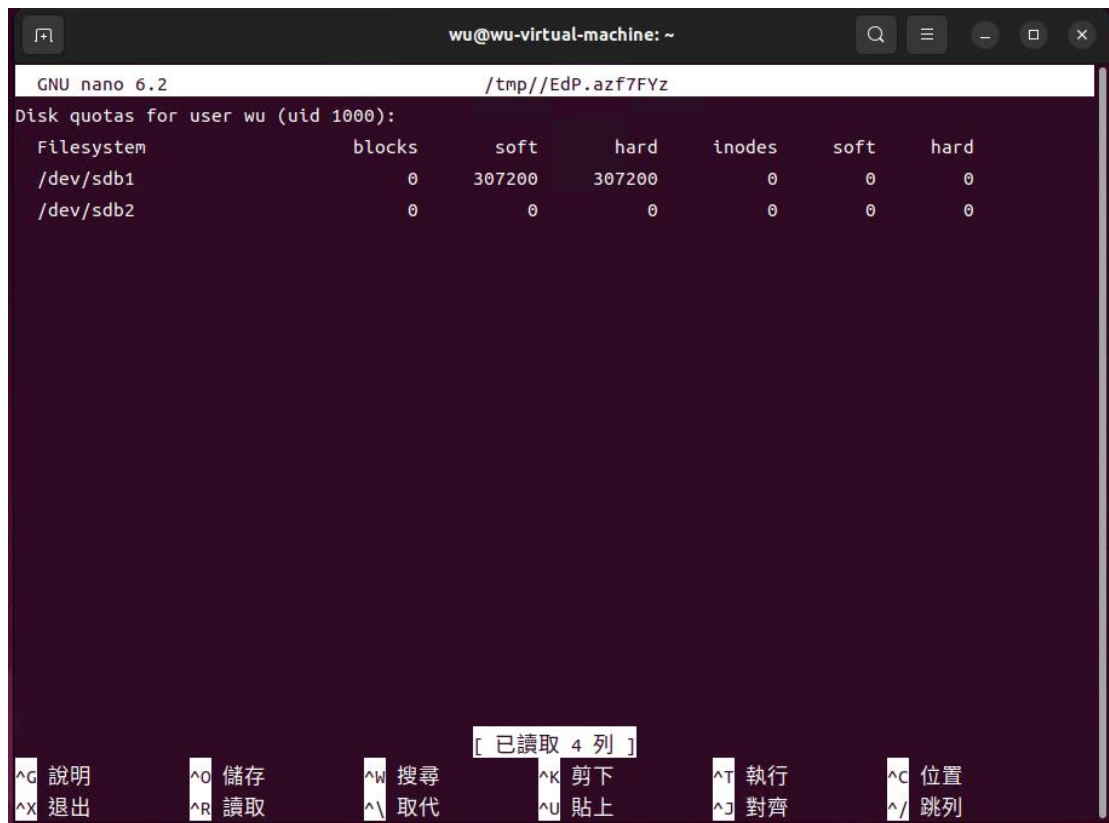
and remount it

sudo mount -o remount /hw

after mount

sudo quotacheck -cug /hw

sudo edquota -u wu



The screenshot shows a terminal window titled 'wu@wu-virtual-machine: ~'. The user is in the GNU nano 6.2 editor, editing the file '/tmp//EdP.azf7FYz'. The output of the 'edquota' command for user 'wu' (uid 1000) is displayed. It shows disk quotas for two filesystems: /dev/sdb1 and /dev/sdb2. The quotas are as follows:

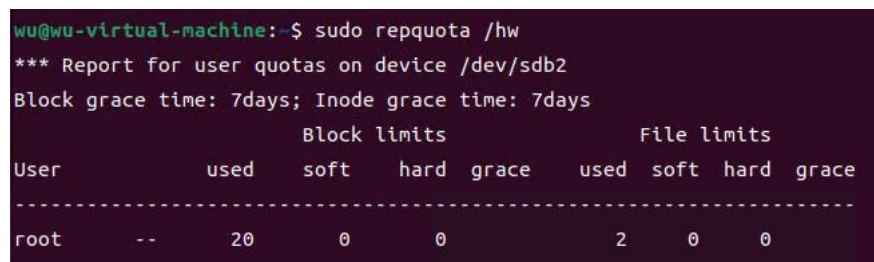
Filesystem	blocks	soft	hard	inodes	soft	hard
/dev/sdb1	0	307200	307200	0	0	0
/dev/sdb2	0	0	0	0	0	0

At the bottom of the terminal, there is a status bar with various keyboard shortcuts and their corresponding actions in Chinese:

- 說明 (Help)
- 儲存 (Save)
- 搜尋 (Search)
- 剪下 (Cut)
- 執行 (Execute)
- 位置 (Position)
- 退出 (Exit)
- 讀取 (Read)
- 取代 (Replace)
- 貼上 (Paste)
- 對齊 (Align)
- 跳列 (Jump)

final report it, use repquota to check actual usage

repquota /hw



The screenshot shows a terminal window titled 'wu@wu-virtual-machine:~'. The user has run the command 'sudo repquota /hw'. The output is a report for user quotas on device /dev/sdb2. It shows the block grace time and inode grace time, both set to 7days. The report then displays the block and file limits for the user 'root'.

User	used	Block limits			grace	File limits		
		used	soft	hard		used	soft	hard
root	--	20	0	0	2	0	0	

Hw_6

I have a set of network domains: 192.168.0.0/28.

What are the Network, Netmask, and Broadcast values of this domain?

What is the number and range of IPs that can be used?

How to observe the routing information (route table) on the Linux host?

What do the SYN and ACK flags on TCP packets mean?

Network, Netmask, and Broadcast values for the subnet 192.168.0.0/28:

Network Address: 192.168.0.0

Netmask: 255.255.255.240 (or /28 in CIDR notation)

Broadcast Address: 192.168.0.15

Number and range of IPs that can be used:

In a /28 subnet (28 consecutive 1's), there are 16 IP addresses in total. However, two of them are reserved (the network and broadcast addresses). So, you can use 14 IP addresses for hosts. The usable IP range is from 192.168.0.1 to 192.168.0.14.

How to observe the routing information (route table) on a Linux host:

You can use the route command or the newer ip route command to view the routing table on a Linux host. Here's how to use them:

route -n displays the routing table.

ip route show provides more detailed information about the routing table.

SYN and ACK flags on TCP packets:

SYN (Synchronize): It is used to initiate a connection. When a client wants to start a TCP connection with a server, it sends a TCP packet with the SYN flag set. This tells the server that the client wants to establish a connection.

ACK (Acknowledgment): It is used to acknowledge the receipt of data. When data is sent in a TCP connection, the receiving end sends ACK packets back to the sender to confirm that it has received the data.

SYN and ACK flags are part of the TCP handshake process used to establish a reliable connection between two devices. The handshake involves a series of SYN, SYN-ACK, and ACK packets exchanged between the client and server to set up the

connection.